

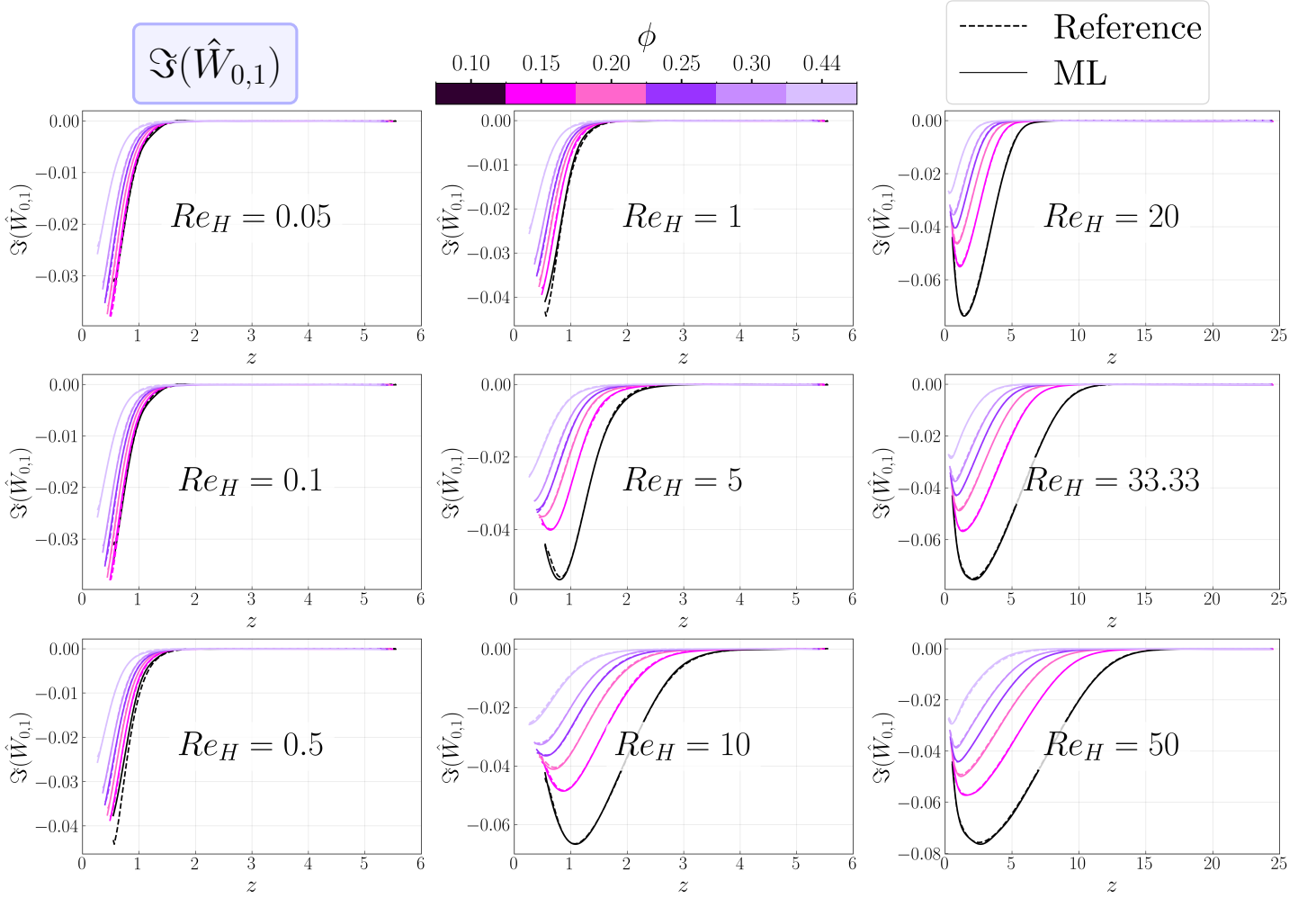


Figure C.3: Comparison of the reference DFT coefficient $\Im(\hat{W}_{0,1})$ (dashed lines) and its neural-network prediction $\Im(\hat{W}_{0,1}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

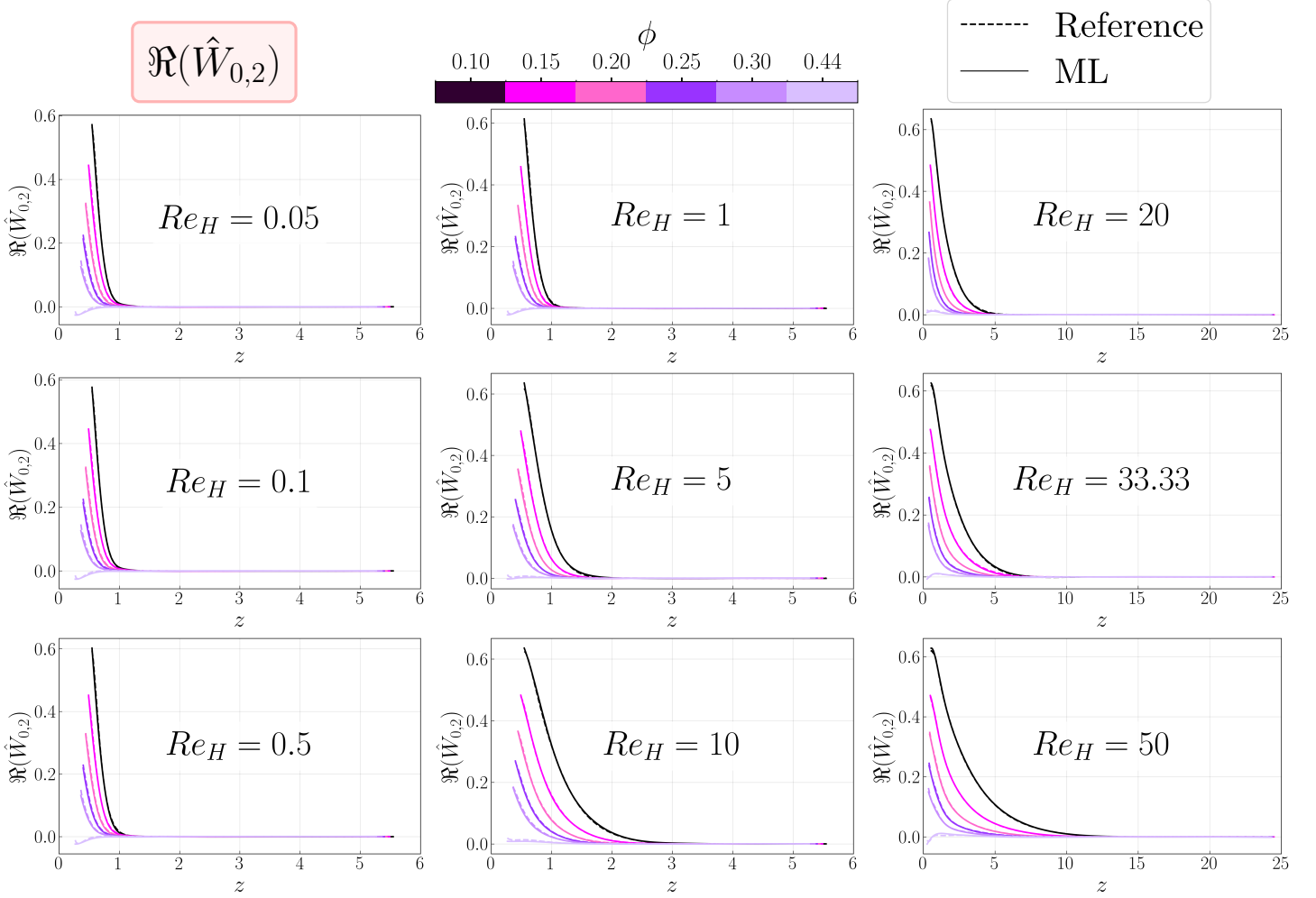


Figure C.4: Comparison of the reference DFT coefficient $\Re(\hat{W}_{0,2})$ (dashed lines) and its neural-network prediction $\Re(\hat{W}_{0,2}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

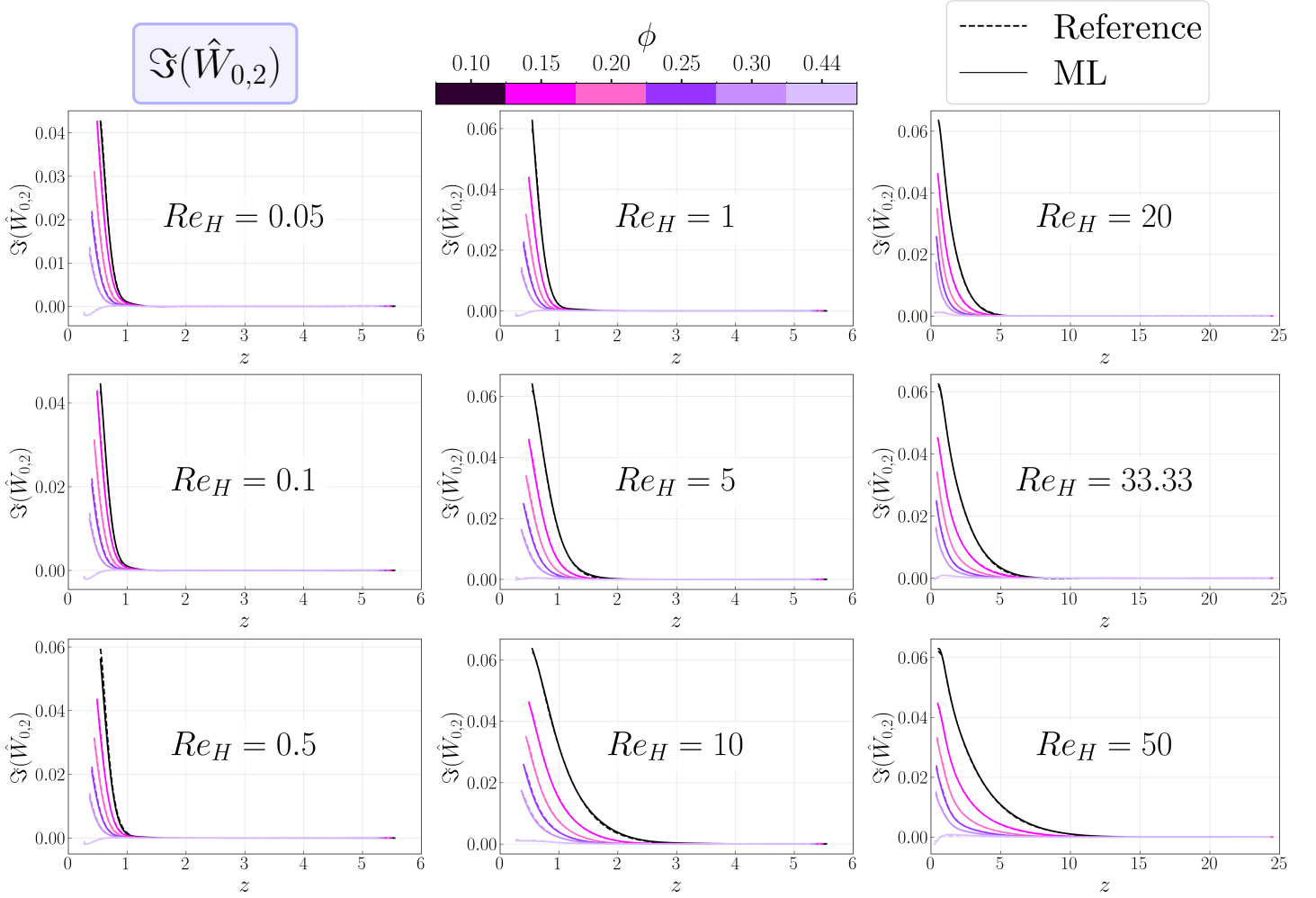


Figure C.5: Comparison of the reference DFT coefficient $\Im(\hat{W}_{0,2})$ (dashed lines) and its neural-network prediction $\Im(\hat{W}_{0,2}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

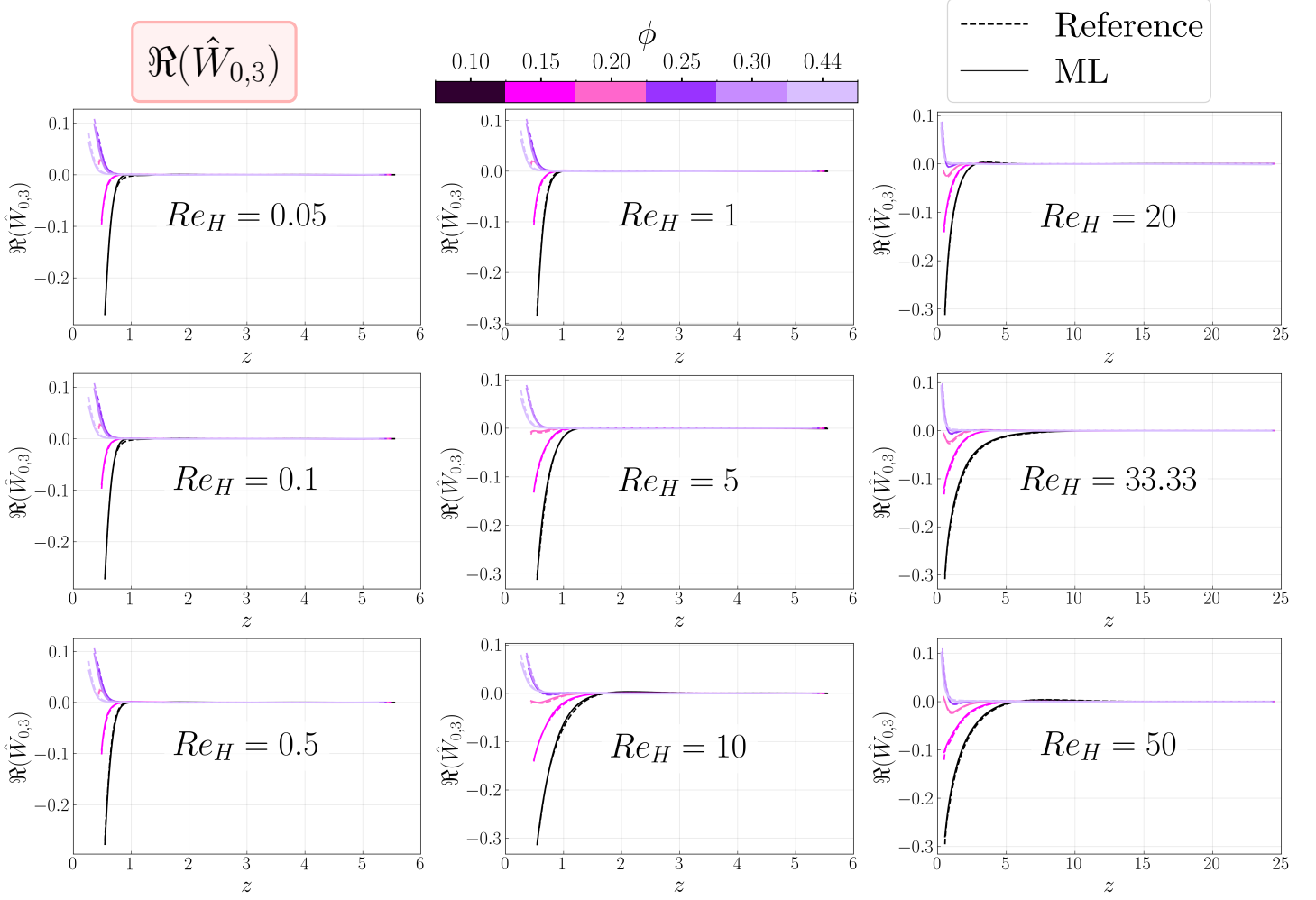


Figure C.6: Comparison of the reference DFT coefficient $\Re(\hat{W}_{0,3})$ (dashed lines) and its neural-network prediction $\Re(\hat{W}_{0,3}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

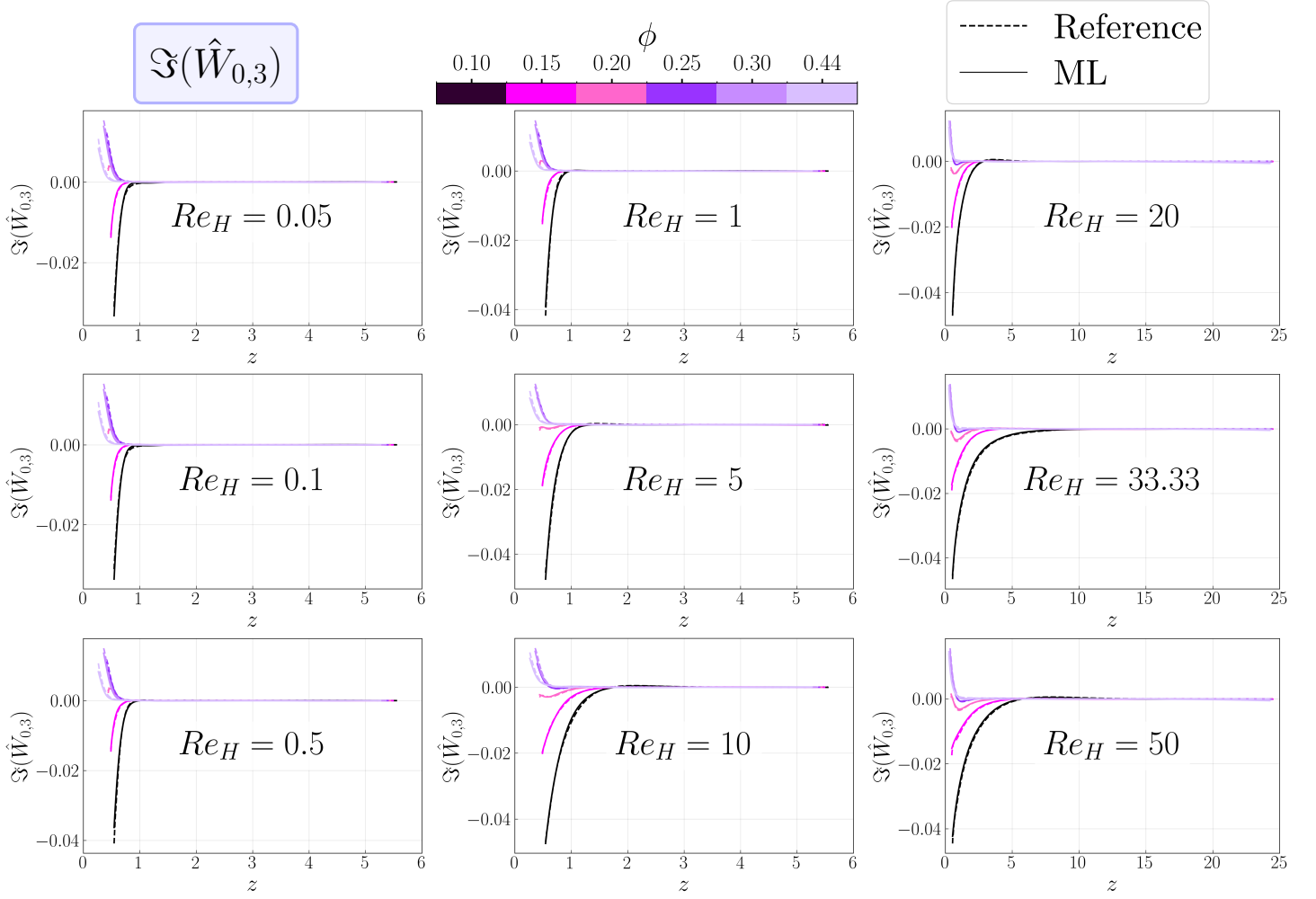


Figure C.7: Comparison of the reference DFT coefficient $\Im(\hat{W}_{0,3})$ (dashed lines) and its neural-network prediction $\Im(\hat{W}_{0,3}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

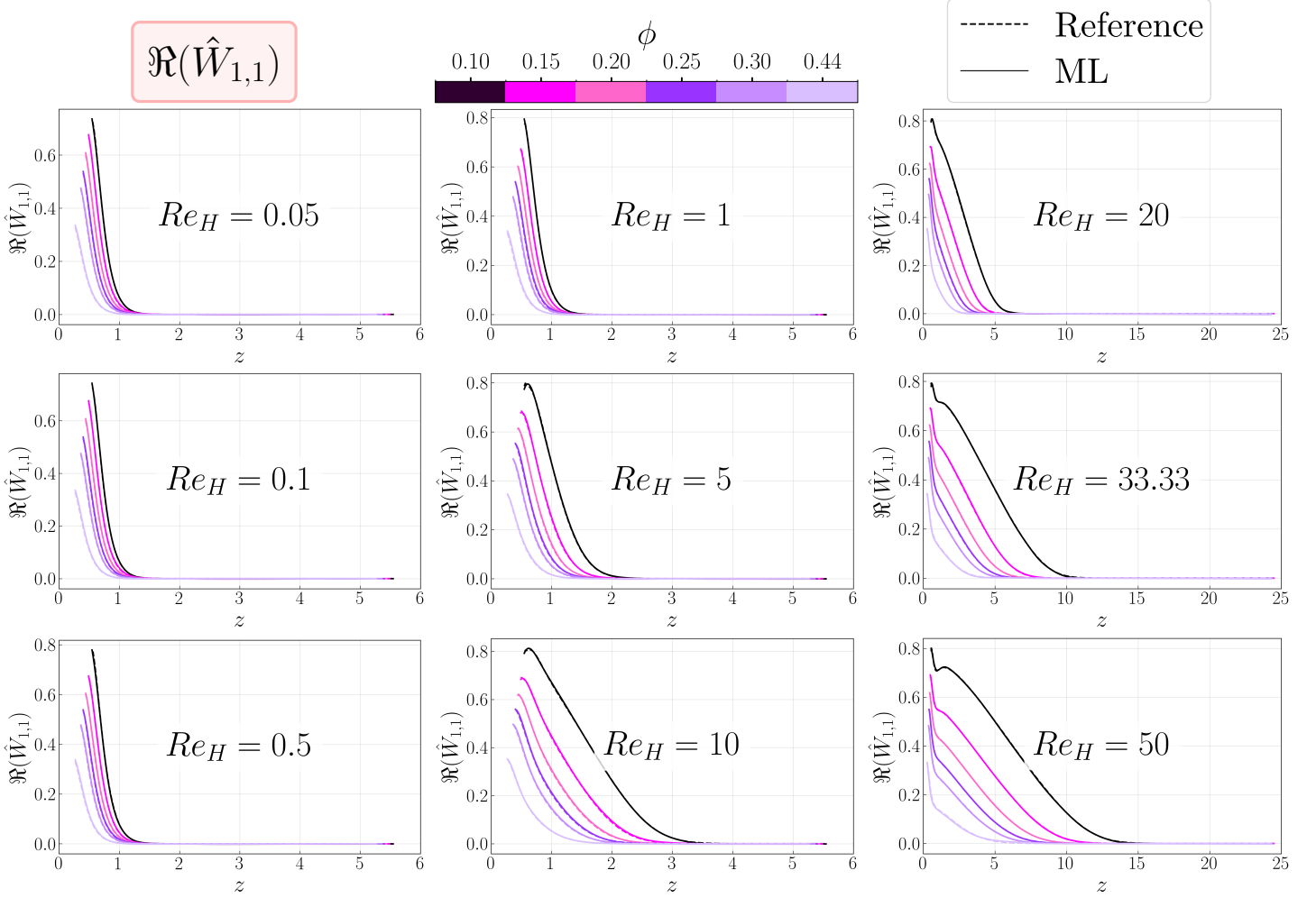


Figure C.8: Comparison of the reference DFT coefficient $\Re(\hat{W}_{1,1})$ (dashed lines) and its neural-network prediction $\Re(\hat{W}_{1,1}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

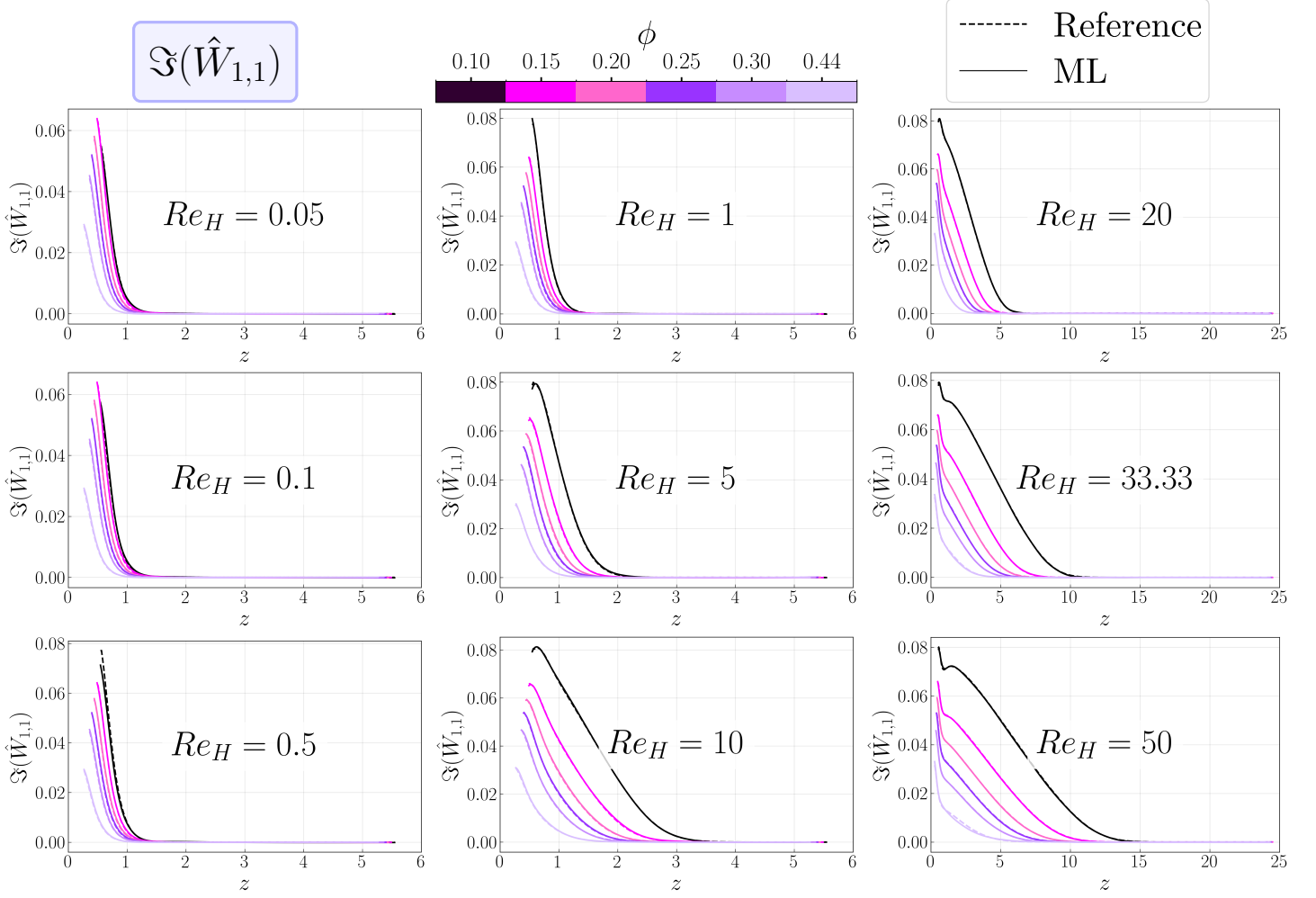


Figure C.9: Comparison of the reference DFT coefficient $\Im(\hat{W}_{1,1})$ (dashed lines) and its neural-network prediction $\Im(\hat{W}_{1,1}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

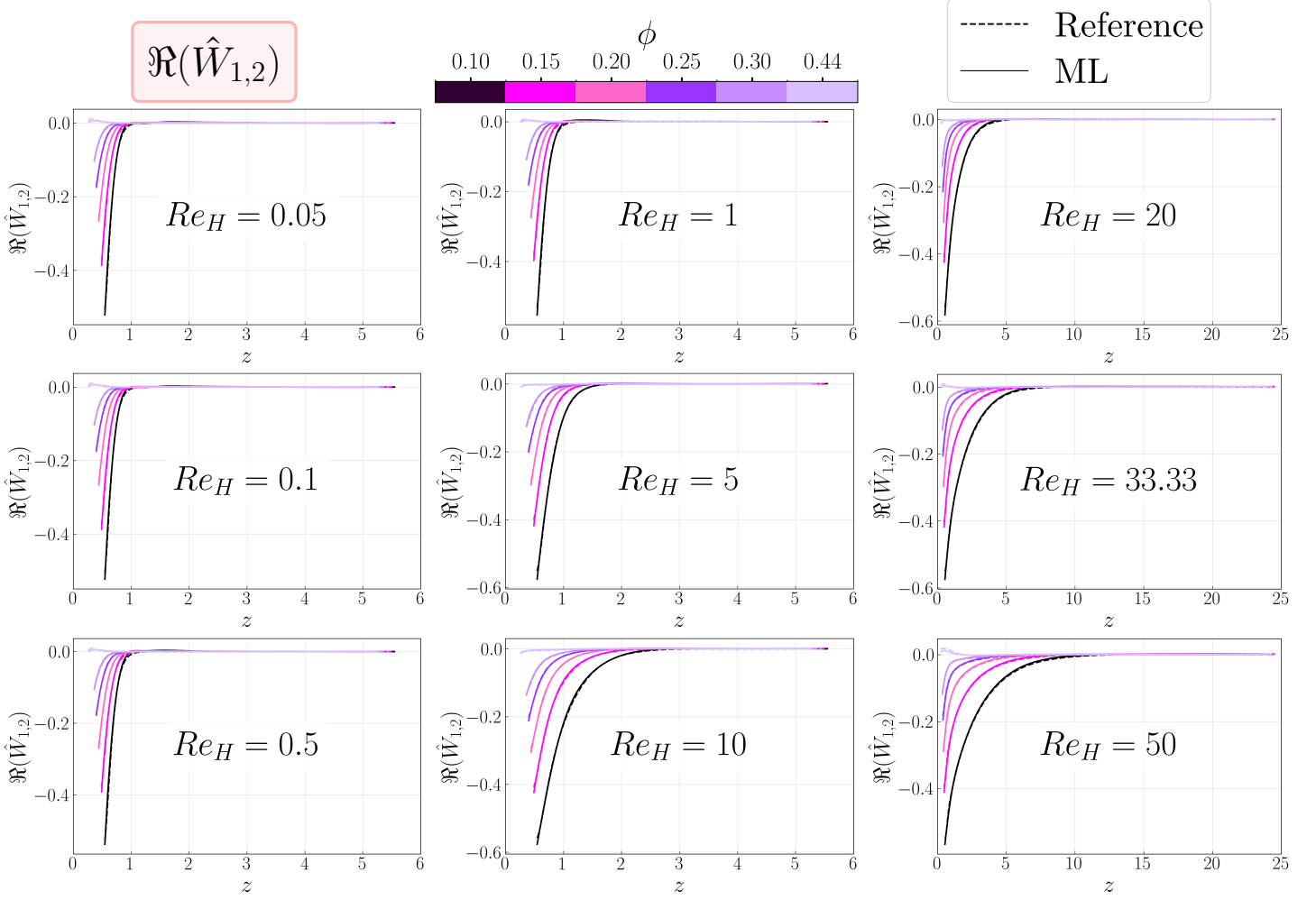


Figure C.10: Comparison of the reference DFT coefficient $\Re(\hat{W}_{1,2})$ (dashed lines) and its neural-network prediction $\Re(\hat{W}_{1,2}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

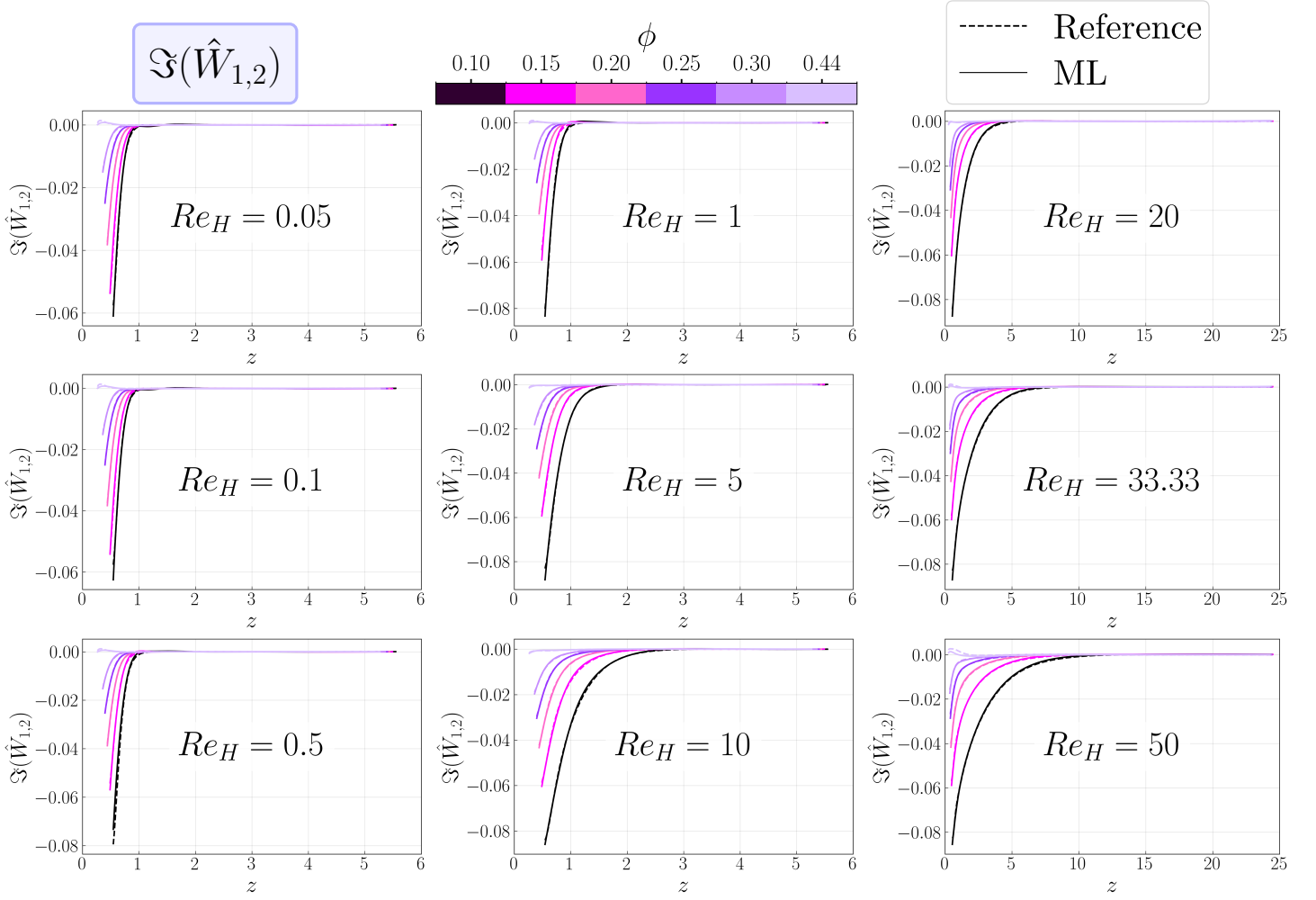


Figure C.11: Comparison of the reference DFT coefficient $\Im(\hat{W}_{1,2})$ (dashed lines) and its neural-network prediction $\Im(\hat{W}_{1,2}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

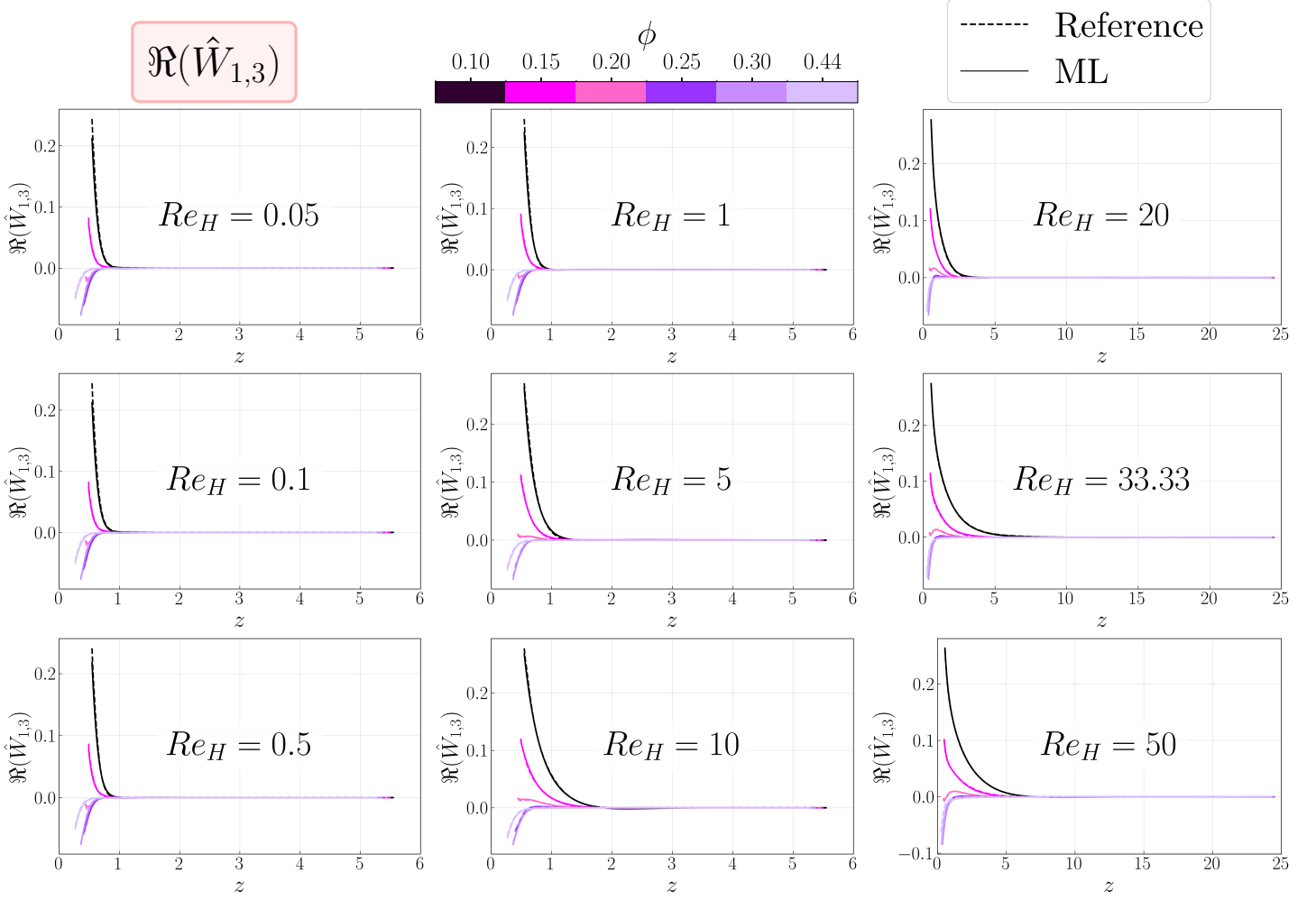


Figure C.12: Comparison of the reference DFT coefficient $\Re(\hat{W}_{1,3})$ (dashed lines) and its neural-network prediction $\Re(\hat{W}_{1,3}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

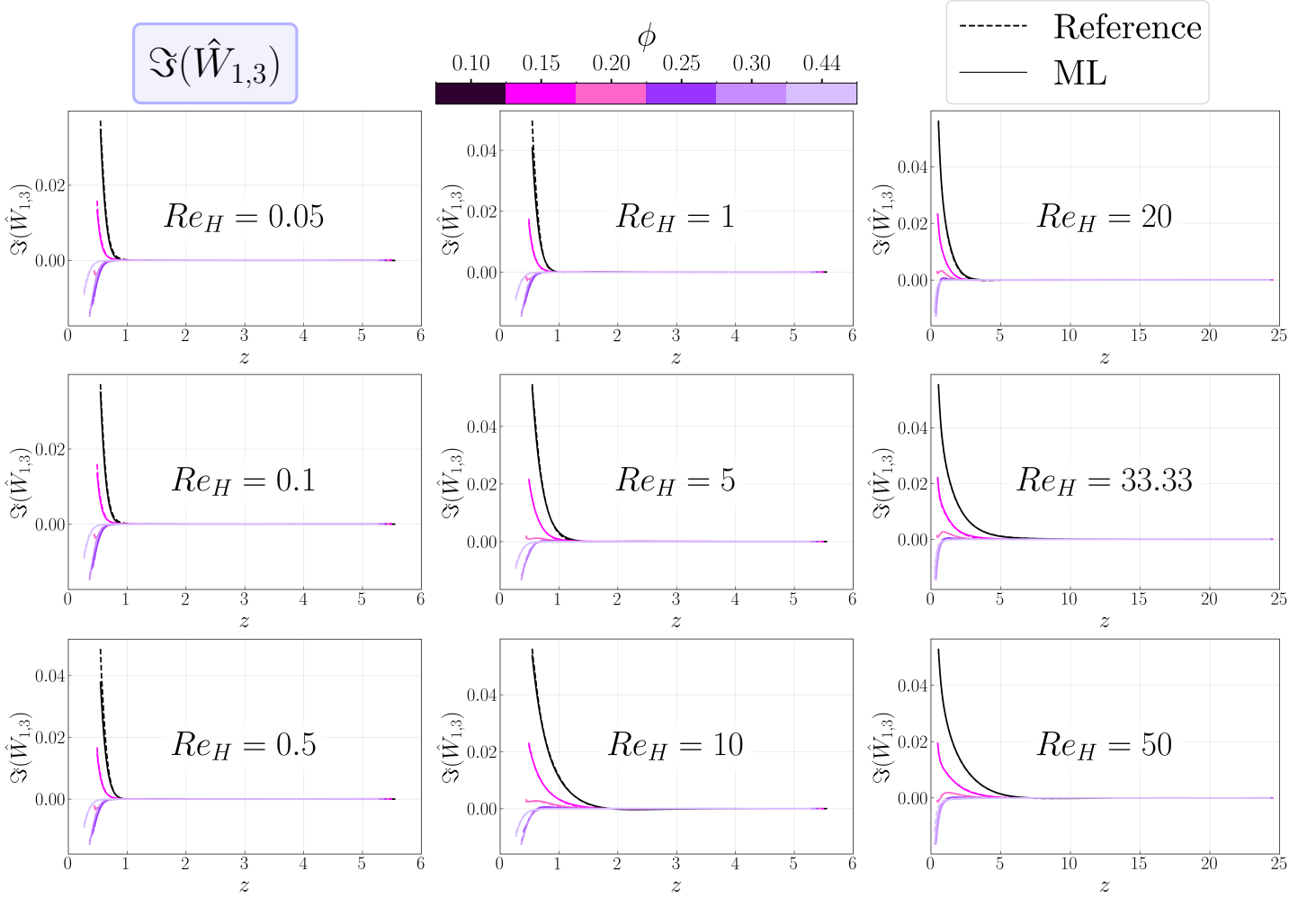


Figure C.13: Comparison of the reference DFT coefficient $\Im(\hat{W}_{1,3})$ (dashed lines) and its neural-network prediction $\Im(\hat{W}_{1,3}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

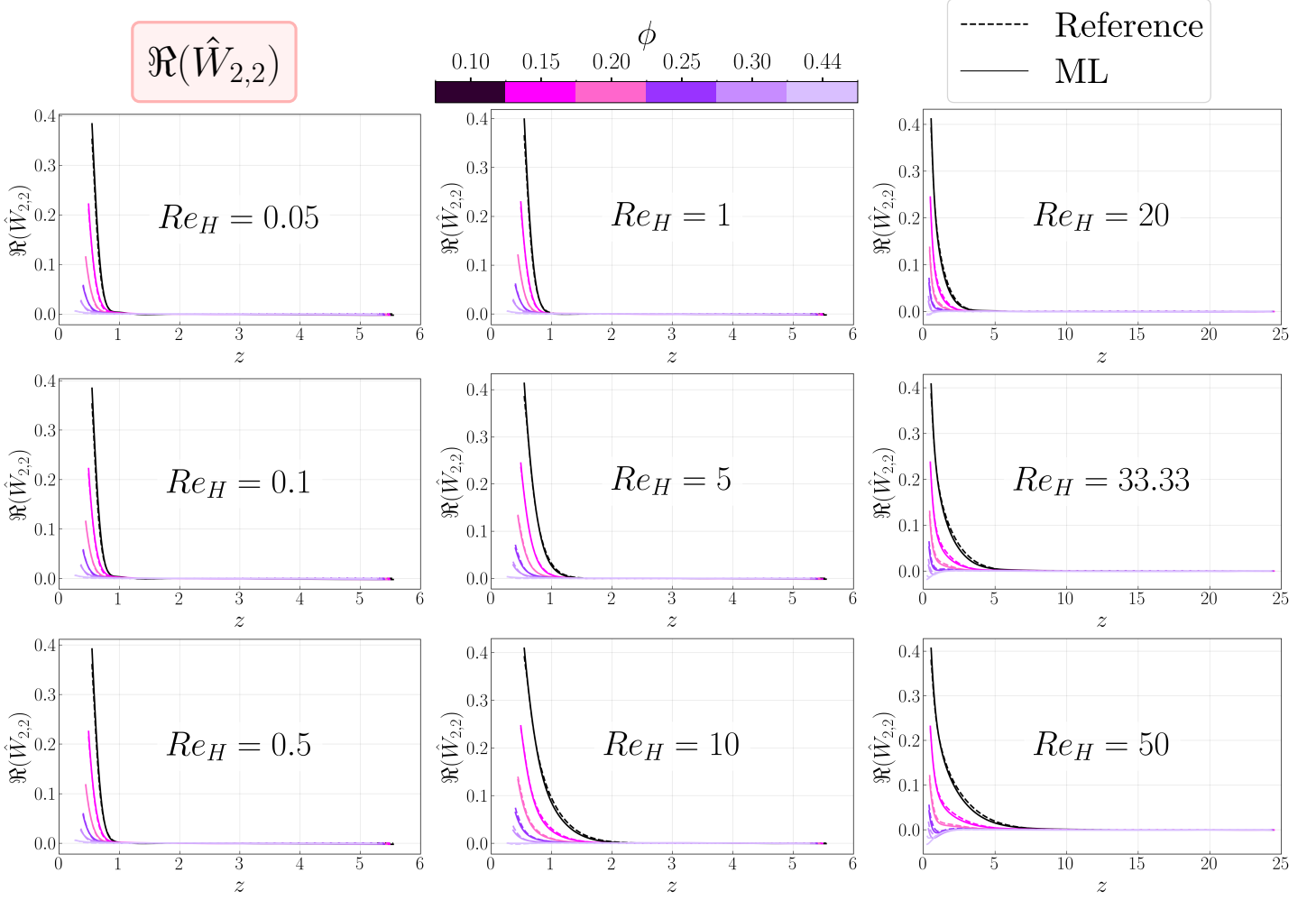


Figure C.14: Comparison of the reference DFT coefficient $\Re(\hat{W}_{2,2})$ (dashed lines) and its neural-network prediction $\Re(\hat{W}_{2,2}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

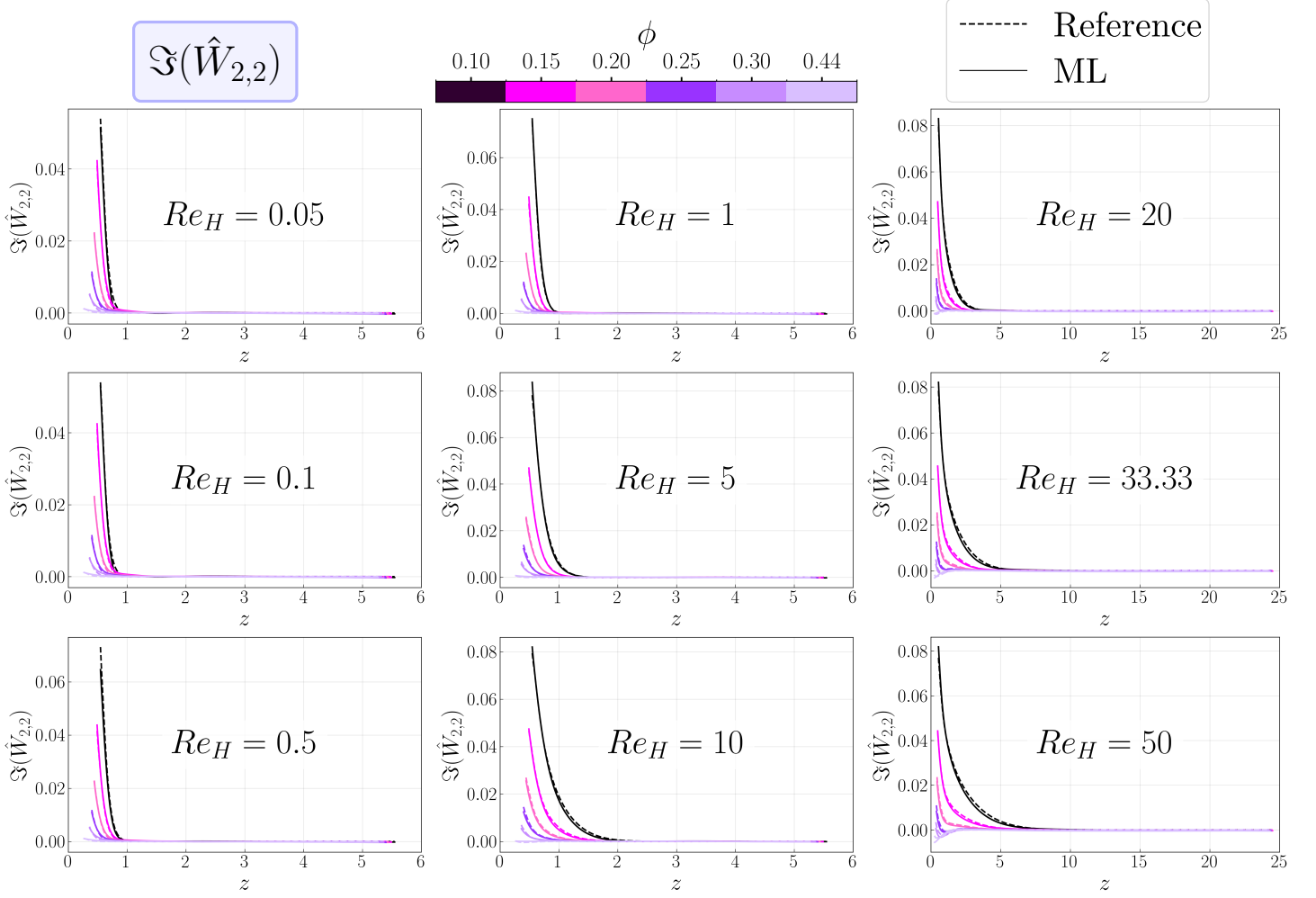


Figure C.15: Comparison of the reference DFT coefficient $\Im(\hat{W}_{2,2})$ (dashed lines) and its neural-network prediction $\Im(\hat{W}_{2,2}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

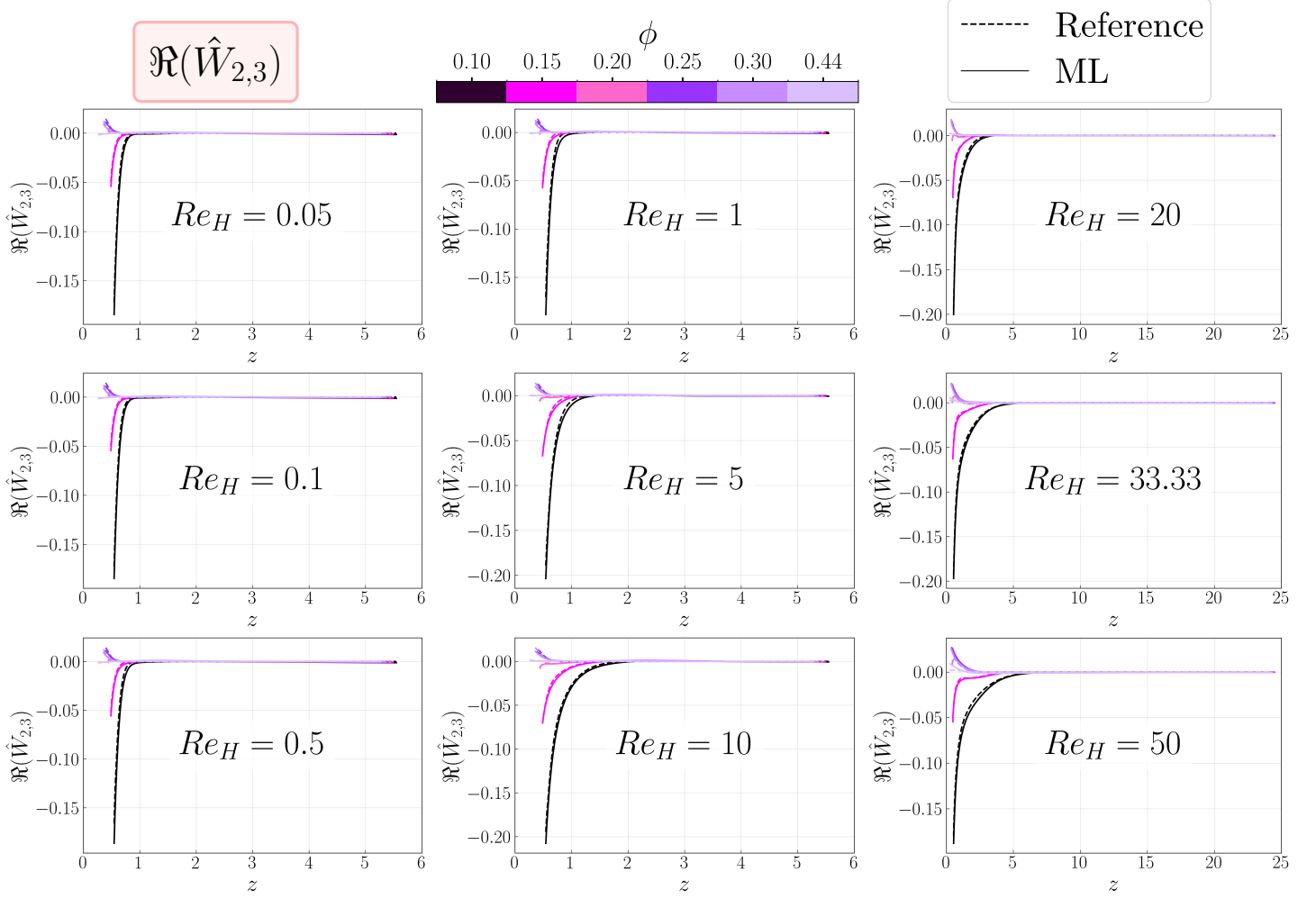


Figure C.16: Comparison of the reference DFT coefficient $\Re(\hat{W}_{2,3})$ (dashed lines) and its neural-network prediction $\Re(\hat{W}_{2,3}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

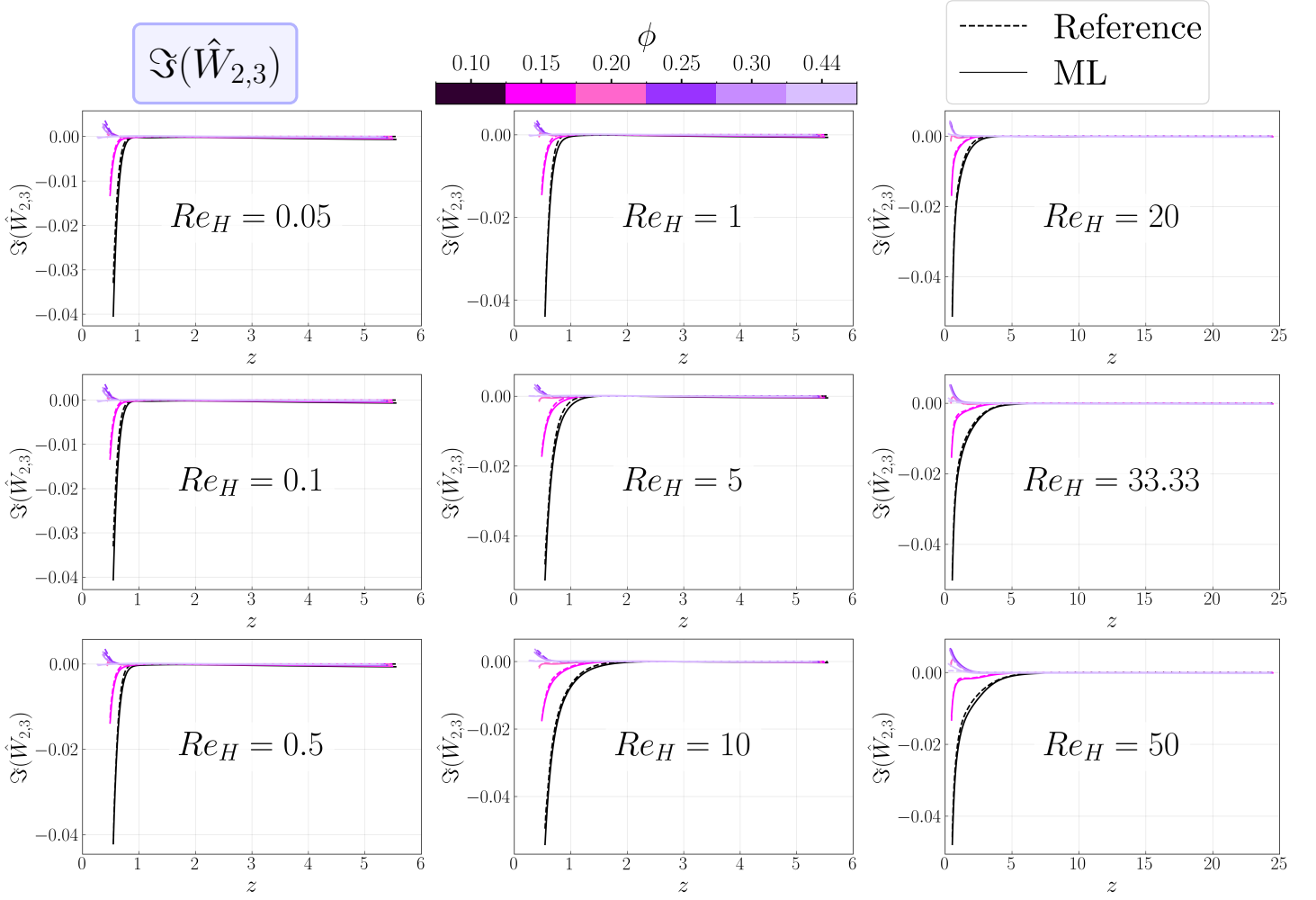


Figure C.17: Comparison of the reference DFT coefficient $\Im(\hat{W}_{2,3})$ (dashed lines) and its neural-network prediction $\Im(\hat{W}_{2,3}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

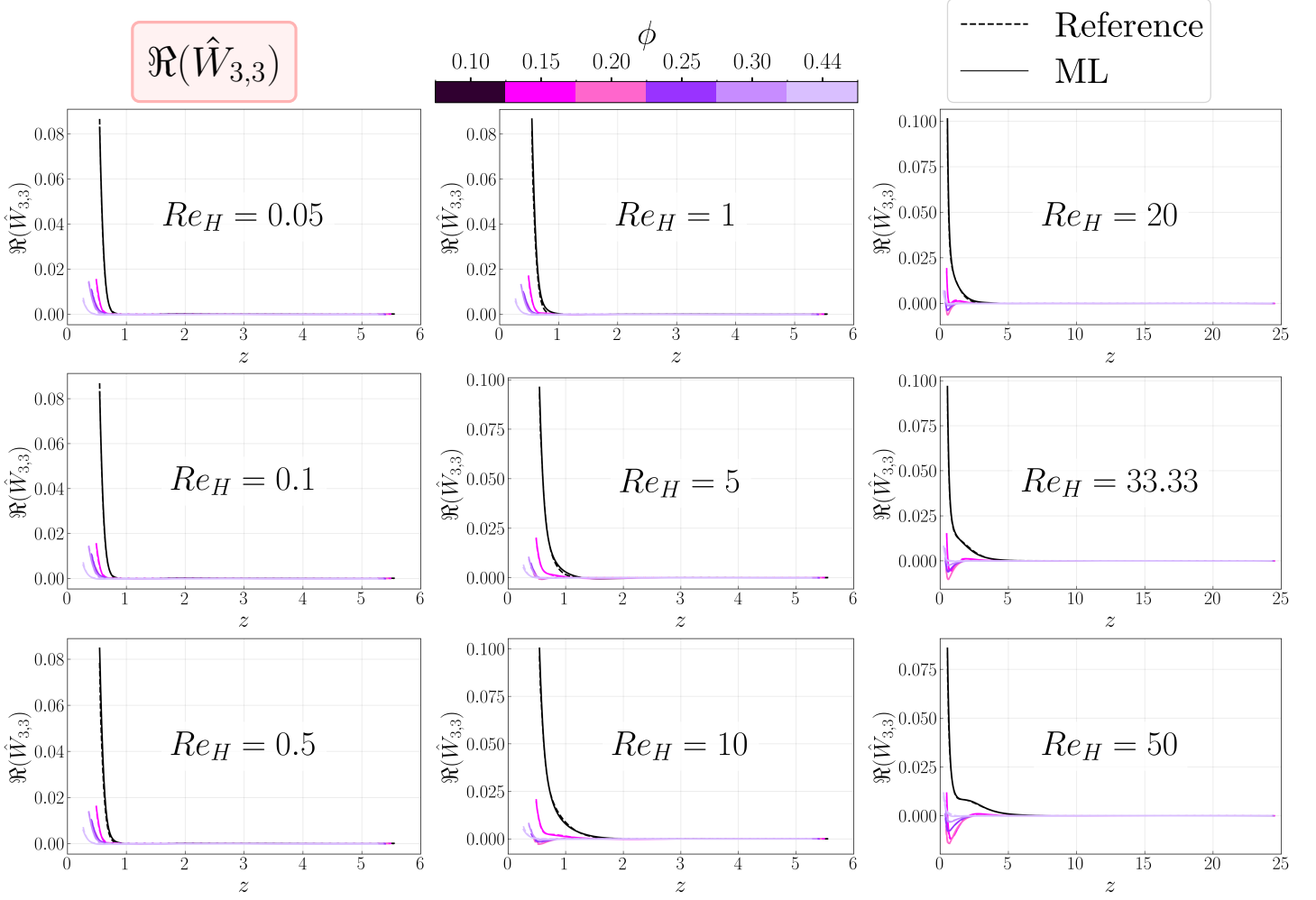


Figure C.18: Comparison of the reference DFT coefficient $\Re(\hat{W}_{3,3})$ (dashed lines) and its neural-network prediction $\Re(\hat{W}_{3,3}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

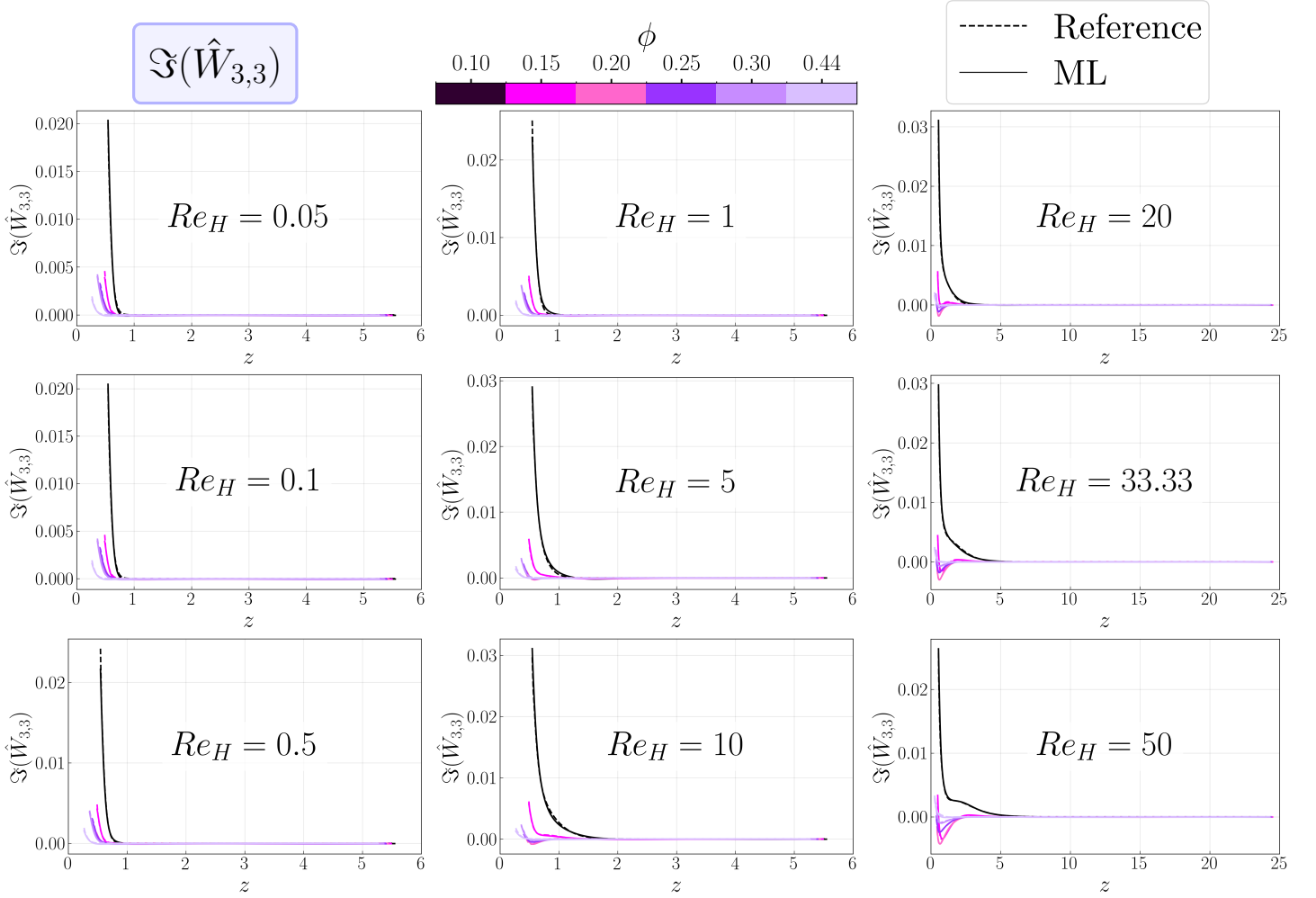


Figure C.19: Comparison of the reference DFT coefficient $\Im(\hat{W}_{3,3})$ (dashed lines) and its neural-network prediction $\Im(\hat{W}_{3,3}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

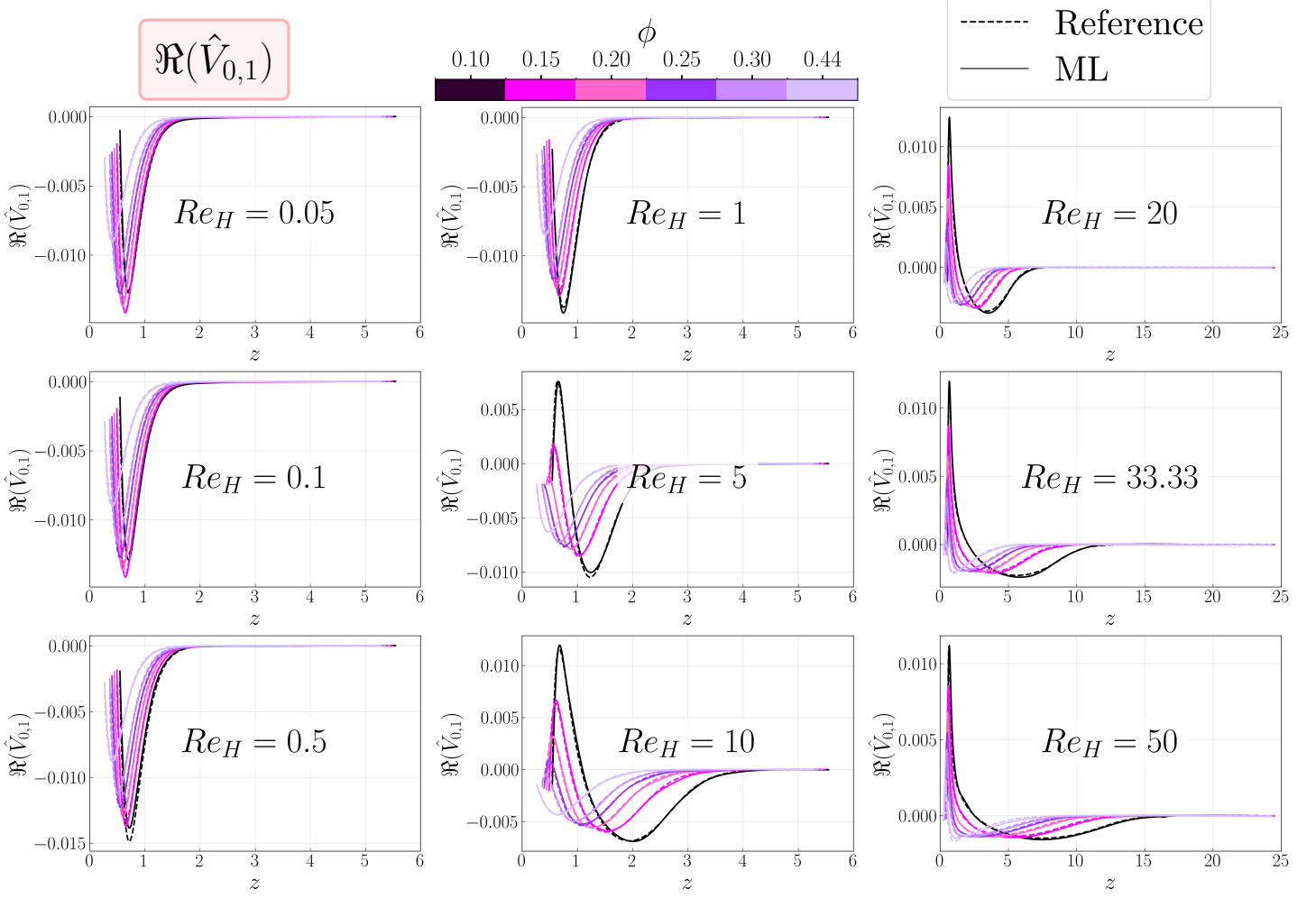


Figure C.20: Comparison of the reference DFT coefficient $\Re(\hat{V}_{0,1})$ (dashed lines) and its neural-network prediction $\Re(\hat{V}_{0,1}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

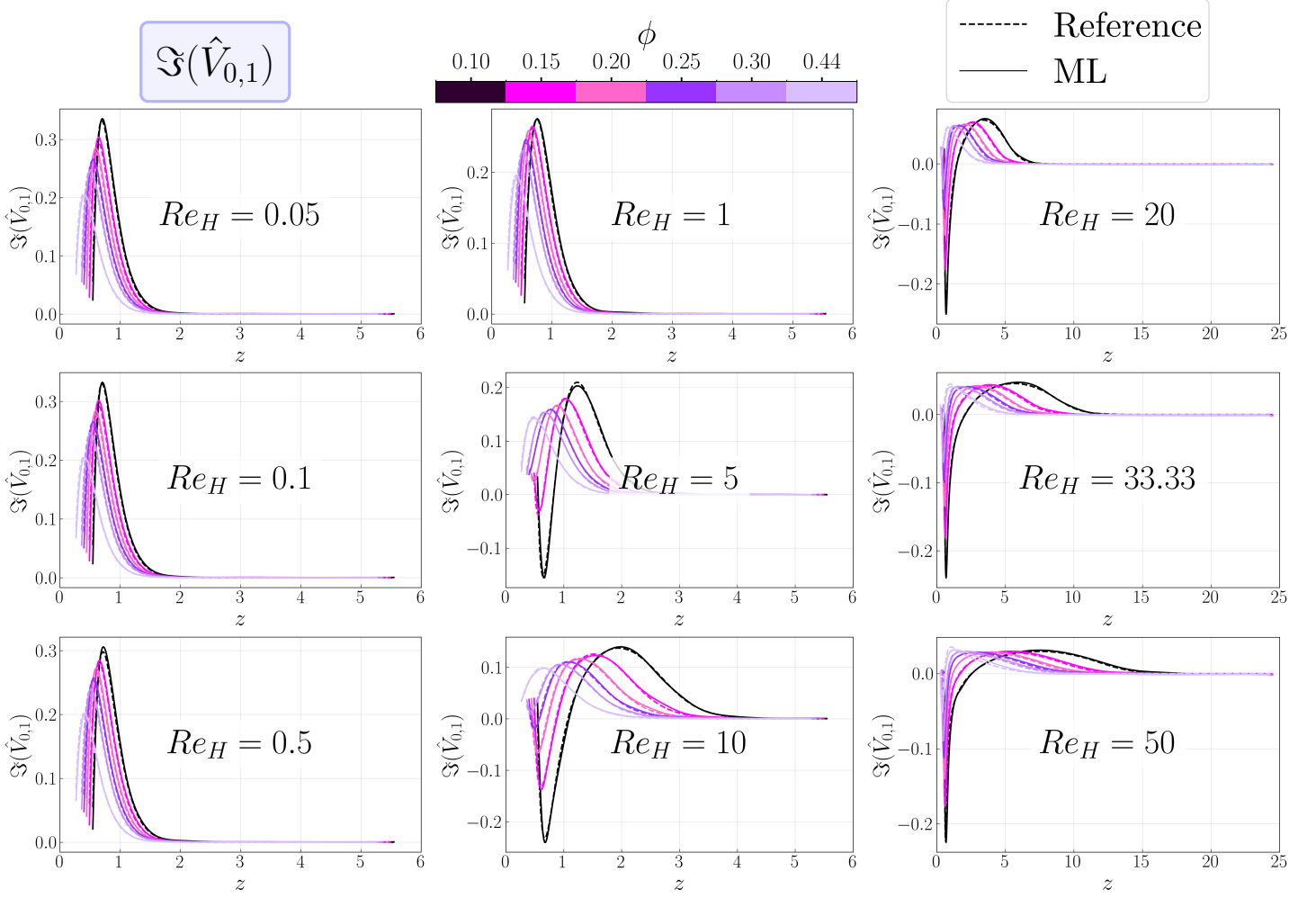


Figure C.21: Comparison of the reference DFT coefficient $\Im(\hat{V}_{0,1})$ (dashed lines) and its neural-network prediction $\Im(\hat{V}_{0,1}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

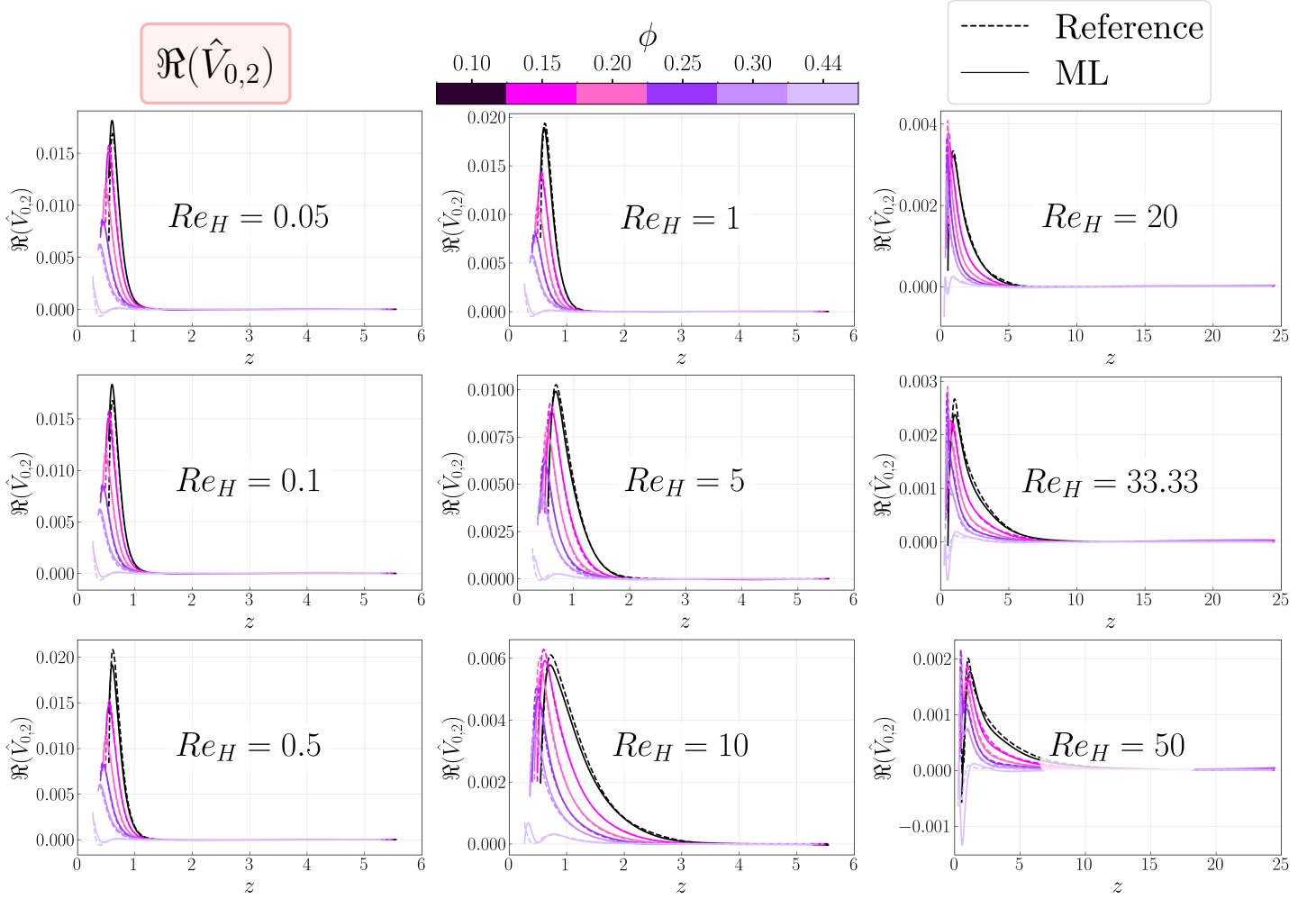


Figure C.22: Comparison of the reference DFT coefficient $\Re(\hat{V}_{0,2})$ (dashed lines) and its neural-network prediction $\Re(\hat{V}_{0,2}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

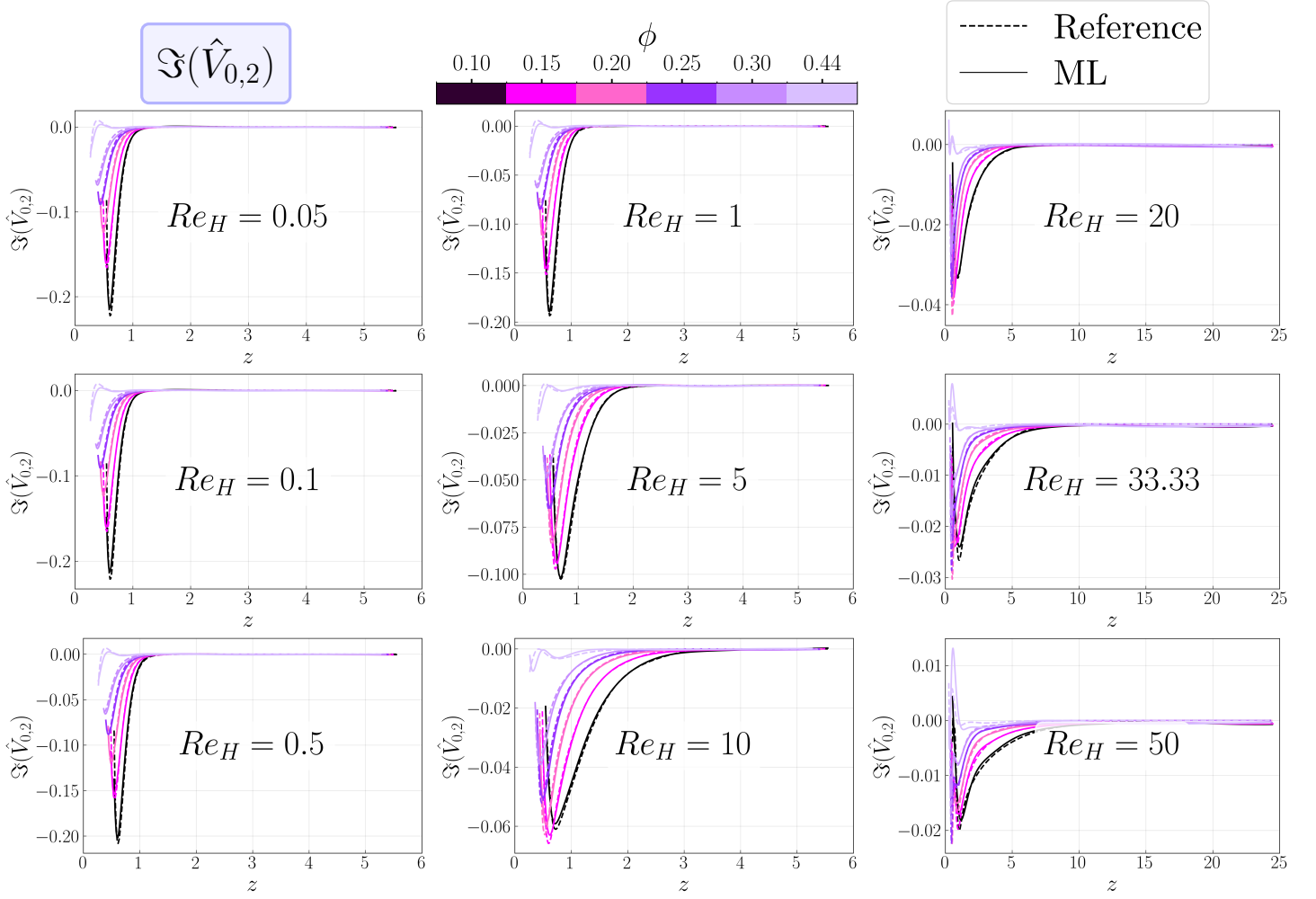


Figure C.23: Comparison of the reference DFT coefficient $\Im(\hat{V}_{0,2})$ (dashed lines) and its neural-network prediction $\Im(\hat{V}_{0,2}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.